The c_J versus Σ threshold

den **is exact; the naive Σ-bound is loose; the reserve shrinks**

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Abstract

Round 419 reduced cofinal vacuous sch < 0 to $\varphi_{bb} < 0$, i.e. $c_J + \Sigma < 0$. The border side $c_J = \mathrm{den} - 2$ is an explicit source sum. The bulk side $\Sigma = 2s^T(C+I)(C-I)^{-1}s$ lives on moderate C-modes ($s \perp C_{\min}$ at 10^{-10}). The hoped conditional theorem — two explicit source sums with an $O(1)$ gap — is **refuted**: den < 2 has an $O(1)$ gap, but the *balance* reserve $-\varphi_{bb}$ is $O(0.03)$ and *shrinks* along the selected sequence. No RH claim.

Status. Lemma 1 is **reduced**. The den formula is a finite identity (Proposition 1). An $O(1)$ source-pure gap $c_J+\Sigma \leq -\delta$ is refuted. Ausgang `DEN_EXACT / BOUND_LOOSE / RESERVE_SHRINKS / COFINAL_OPEN`. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

Lemma 1 (c_J vs Σ). *On the vacuous branch at large N, $c_J < -\Sigma$ (equivalently $\varphi_{bb} < 0$, equivalently* sch < 0 *cofinally via the r419 asymptotics), source-pure.*

Verdict: REDUZIERT. Selected $k = 4, \ldots, 9$ and EXT vacuous windows have $c_J < -\Sigma$ as a census. The reserve shrinks. No k_0, no tight source-pure upper bound on Σ.

2 The den formula

Proposition 1 (border normalisation). *Write $B_w = S_{N-2} + \frac{5}{7}$ for the r362/r397 budget ($S = \sum \rho_k$, $\rho_k = F_k^2/h_k$ the von Mangoldt border-chain masses) and b the μ-orthonormal border vector. Let $v = B_\mu b/\sqrt{B_w}$, $\gamma = \|b\|^2/B_w$, $v_t = \varepsilon_Y \odot v$ and $s = R_{N-3}v_t$. Then*

$$\mathrm{den} \;=\; 1 + \gamma - v_t \cdot s, \qquad c_J = \mathrm{den} - 2. \tag{1}$$

Over $\mathbb{Q}$, $\gamma = \frac{2}{3}$, $v_t \cdot s = \frac{1}{6}$ gives den $= \frac{3}{2}$, $c_J = -\frac{1}{2}$. *At $k_z = 9$,* den $= 1.60111$, $\gamma = 0.6778$, $B_w = 8.382$. *On the named census every window has* den $\in [1.460, 1.652] < 2$, *so the gap to* 2 *is* $O(1)$ *in* $[0.348, 0.540]$. *Vacuous* den $\sim N$ *slope* -0.03 *(flat).*

Thus den < 2 is an explicit budget statement with an $O(1)$ margin. It is *not* the balance $c_J + \Sigma < 0$.

3 Where s sits; the Σ bound

Proposition 2 (occupation and the naive bound). *On vacuous windows s occupies moderate C-modes (mass-weighted $\lambda \sim 3.2$–4.6). The $C_{\min}$-share is $\leq 3 \cdot 10^{-7}$. $\|s\|^2 \sim 0.06$ is flat. The bound $\Sigma \leq 2\|s\|^2 \max_{\mathrm{occ}}(\lambda+1)/(\lambda-1)$ over modes with relative mass $> 10^{-6}$ is* 200–24000 *times Σ: near-razor modes with tiny occupation blow the maximum. No tight source-pure upper bound of the form $\Sigma \leq C_{\mathrm{expl}}\|s\|^2$ with $C_{\mathrm{expl}} = O(1)$ is obtained. Σ itself is $O(0.40)$ and flat.*

4 The reserve; the six; selected

Write $R = -c_J - \Sigma = -\varphi_{bb}$.

Proposition 3 (reserve)**.** *Healthy vacuous core:* $R \in [0.011, 0.068]$*. The overflow six:* $R < 0$ *(they live by* $\tau^2 > \varphi_{bb}$*). EXT vacuous:* $R \in [0.029, 0.037]$*. Selected* $a_k = 2^k$ *at* $k = 4, 5, 6, 7, 9$ *(and* $k = 8$ *in calibration,* $N = 5690$*): all* $R > 0$*. The reserve* shrinks*:* $k = 5$*:* 0.105*;* $k = 6$*:* 0.068*;* $k = 7$*:* 0.052*;* $k = 8$*:* 0.048*;* $k = 9$*:* 0.038*. The hoped* $O(1)$ *gap is refuted.*

The six remain a named finite class. $k = 10$ ($k_z = 197$) was not computed (border pack fail).

5 One balance

Proposition 4 (both charts)**.** $\varphi_{bb} = c_J + \Sigma$ *on P1 and vacuous alike. On living P1,* $R_+/|\Sigma|$ *has median* 1.009 *— almost all of* Σ *is the positive rest.* $25/28$ *P1 windows have* $c_J + R_+ < 0$*;* $28/28$ *have* $c_J + \Sigma < 0$*. The whole edge is one* c_J*-versus-self-energy balance.*

6 Kills

Dead vacuous χ violate *both* sides: $\Sigma > |c_J|$ and $\tau^2 < \varphi_{bb}$. Dropping the -2 in c_J sends $w9$ to $+1.914$. Drop-border: no φ_{bb} scalar. Scramble $\mathrm{nneg}(A_0) = 21$; two-period $S = 21$ has $\mathrm{nneg} = 4$.

7 What is a theorem, what is a census

Proposition 1 is a theorem. Lemma 1 is not: the $O(1)$ gap fails, the naive Σ-bound is loose, and the selected reserve shrinks. Cofinal vacuous $c_J < -\Sigma$ remains a shrinking $O(0.03)$ census, no k_0. No RH claim.